

12.9.2025

CS 310

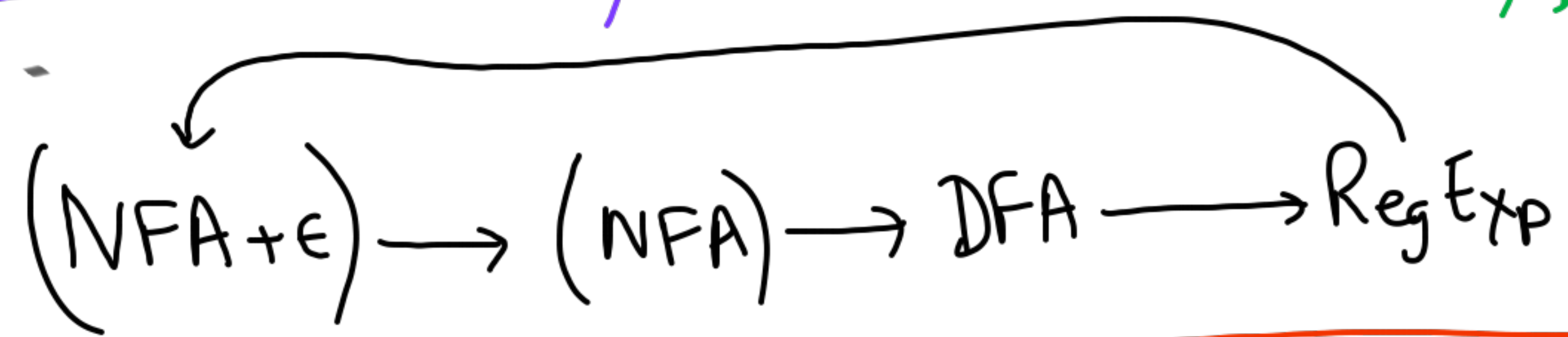
Languages — Machines

Canonical (Normal) Forms

→ Closure Properties
→ Determinism

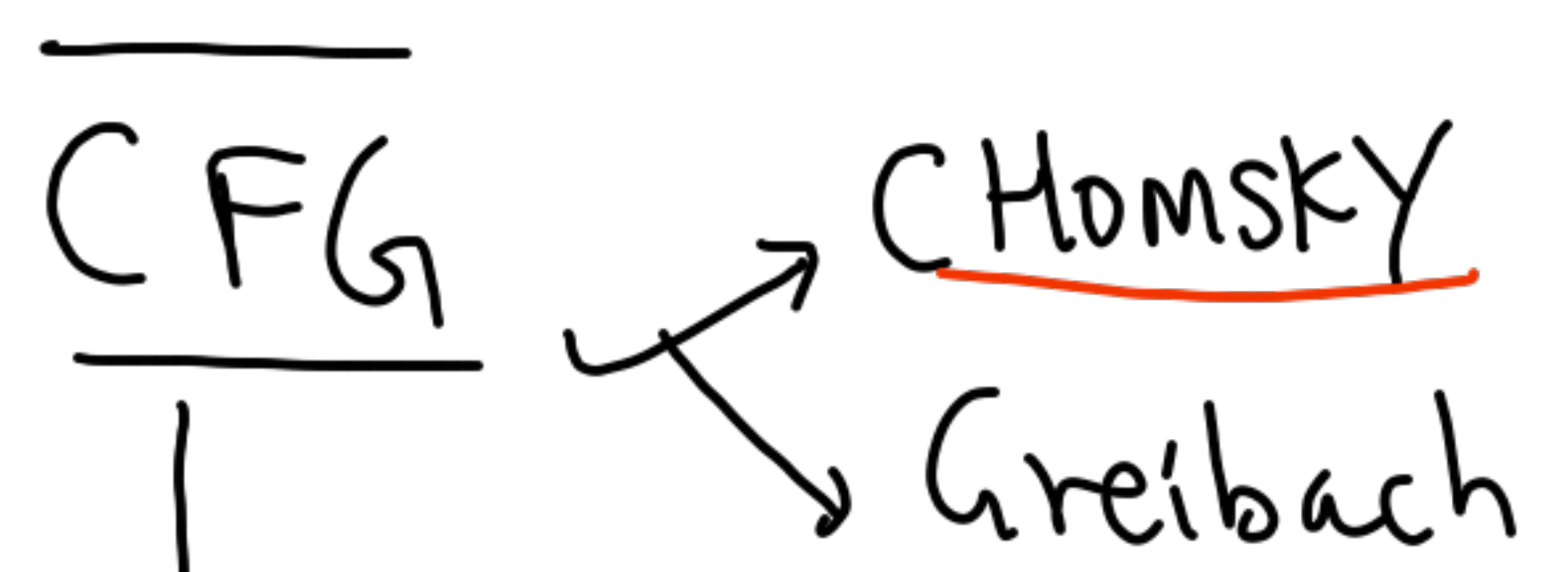
①

10 mins



10 mins

②



50 mins

③

PUMPING Lemma

MIDSEM

→ Sat 20/9

~~LA 01, 02~~
201

Ensemble
Federated

Moodle

NFA + ϵ

NFA

DFA

Reg Exp

ϵ -closure(S_7)

no new state

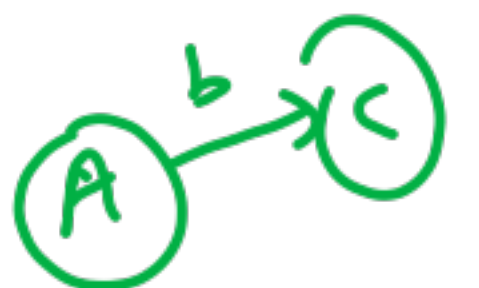
easy

Brute-force

Right Linear Grv

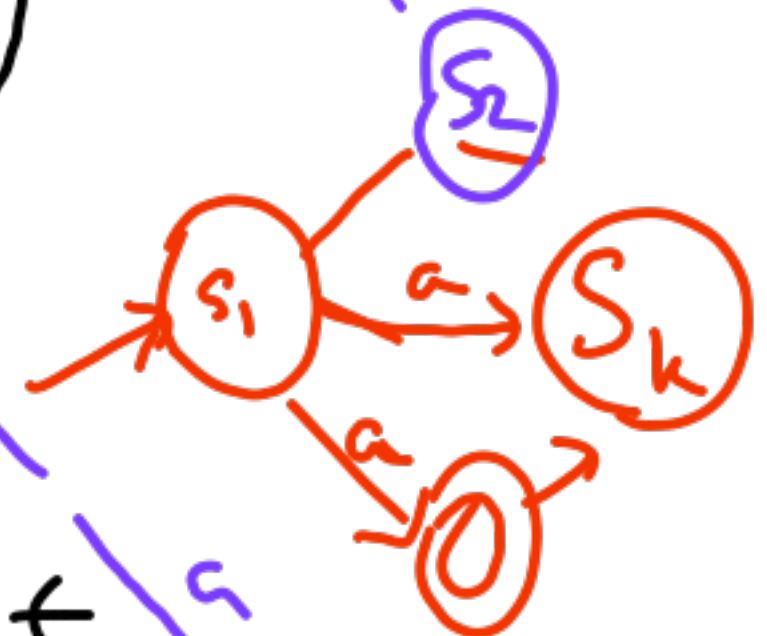
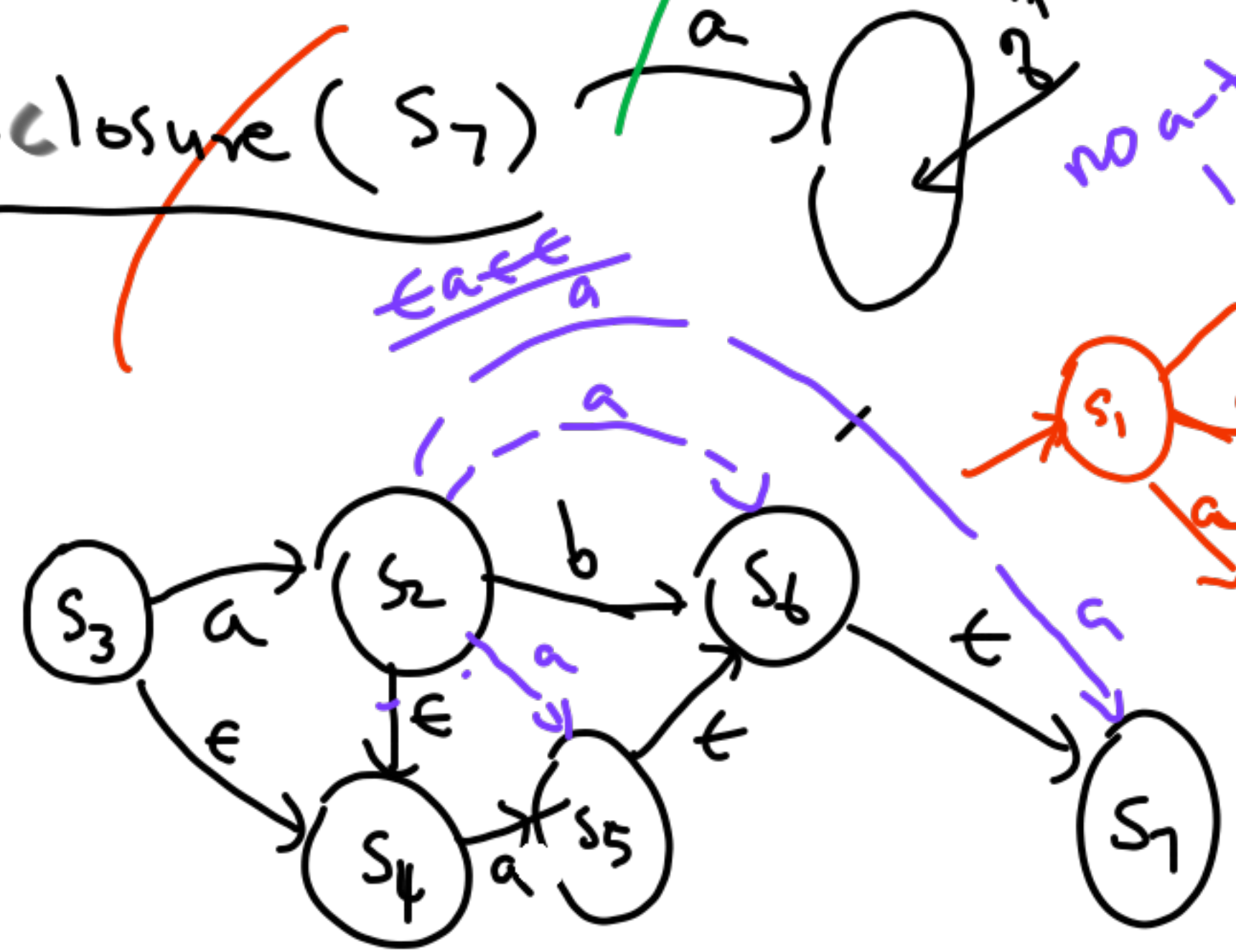
for small examples

Subset 2^R states



$A \rightarrow bC$

Dynamic programming



from $S_3 \xrightarrow{a}$

which states $\xrightarrow{a}$ how many

all answer



Intelligent

generate "as needed"
lazy evaluation

CFG Normal Forms (Σ, V, S, P)

$S \rightarrow ABCD$

Chomsky

$S \rightarrow AX_1$

$X_1 \rightarrow BX_2$

$X_2 \rightarrow CD$

for any $A \in V$

type 2 (T1) $A \rightarrow c$

type 2 (T2) $A \rightarrow BD$

only 2 types of rules

$S \rightarrow \bar{a} S \bar{b}$
 $S \rightarrow \bar{a} b$
 $\bar{x_i} \bar{x_i}$



Sheila Grebach

only 1 type of rule

$A \rightarrow \bar{c} \alpha$ anything

$c \in \Sigma, \alpha \in (V \cup \Sigma)^*$

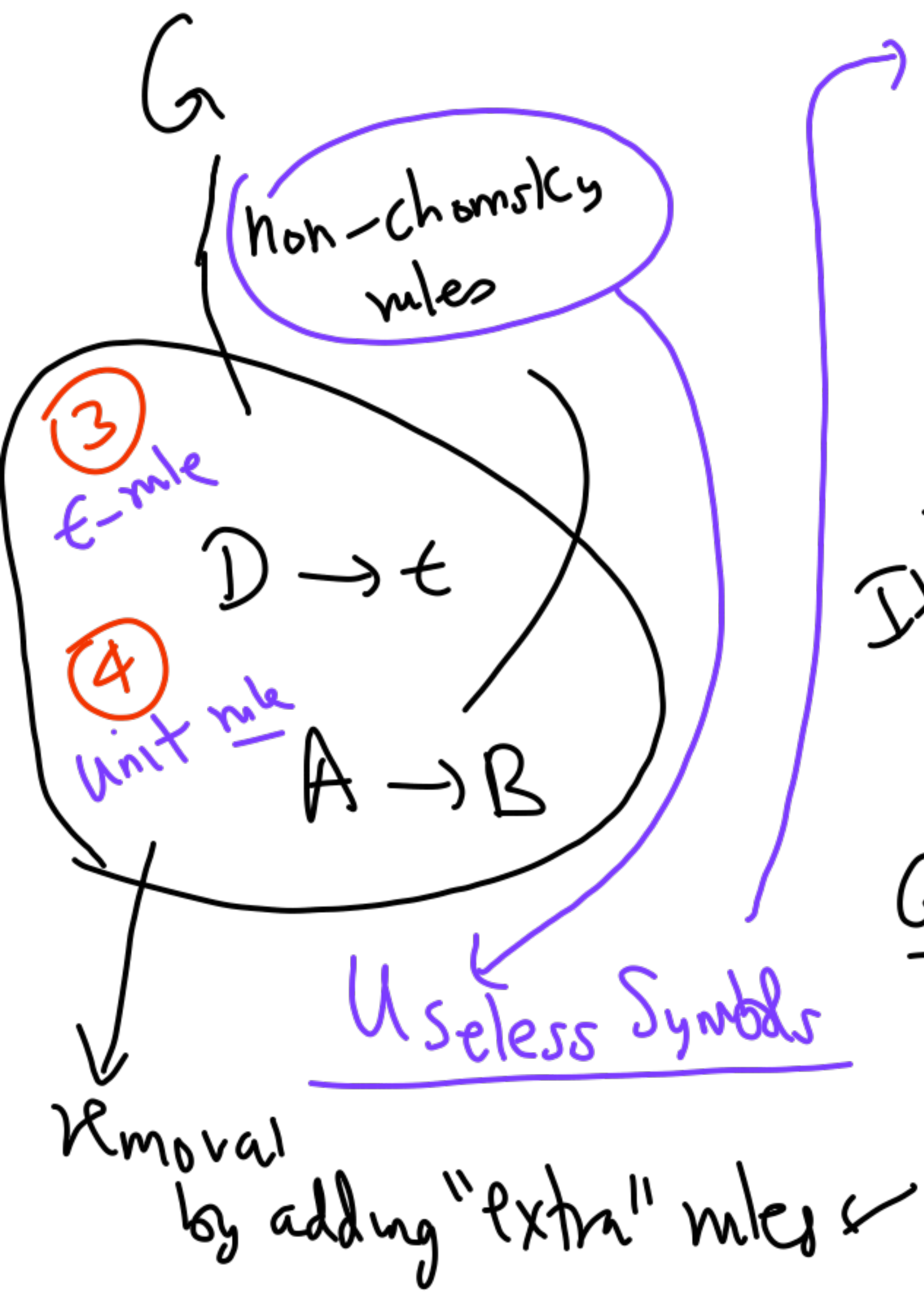
one alphabet

~~$S \rightarrow X_1 S X_2$~~

$S \rightarrow X_1 S_2$

$S_2 \rightarrow S X_2$

$S \rightarrow X_1 X_2$



Useful iff both productive & reachable

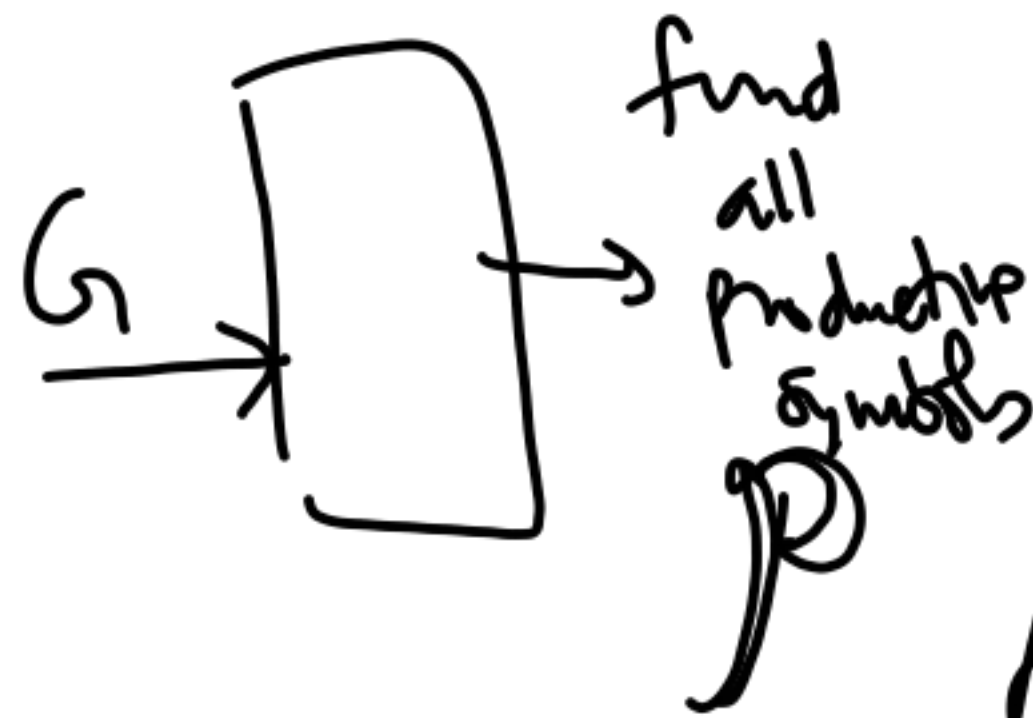
① productive

② reachable

$$A \xrightarrow{*} w \in \Sigma^*$$

$$S \xrightarrow{*} \alpha_L A \alpha_R$$

Iterative Fix-Point Method



① $P_0 = \{ \text{all symbols } D \rightarrow \alpha \in \Sigma^* \}$

② $P_{i+1} = P_i \cup \{ \text{all symbols whose rules have RHS} \in P_i \}$

③ until $P_{i+1} = P_i$

1-step



Pumping lemma

Chomsky
Normal
Form

Pigeonhole

"Some variable
repeats"

Derivation tree

S path
longer
than $|V|$

min depth
to derive
string of
length n
is
 $\log(n)$

